

11.2 Mitigate VLAN Attacks

Mitigate VLAN Attacks

VLAN Attacks Review

A VLAN hopping attack can be launched in one of three ways:

- Spoofing DTP messages from the attacking host to cause the switch to enter trunking mode. From here, the attacker can send traffic tagged with the target VLAN, and the switch then delivers the packets to the destination.
- Introducing a rogue switch and enabling trunking. The attacker can then access all the VLANs on the victim switch from the rogue switch.
- Another type of VLAN hopping attack is a double-tagging (or double-encapsulated) attack. This attack takes advantage of the way hardware on most switches operate.

Steps to Mitigate VLAN Hopping Attacks

Step 1: Disable DTP (auto trunking) negotiations on non-trunking ports by using the **switchport mode access** interface configuration command.

Step 2: Disable unused ports and put them in an unused VLAN.

```
S1(config)# interface range fa0/1 - 16
S1(config-if-range)# switchport mode access
S1(config-if-range)# exit
S1(config)#
S1(config)# interface range fa0/17 - 20
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 1000
S1(config-if-range)# shutdown
S1(config-if-range)# exit
```

Steps to Mitigate VLAN Hopping Attacks

Step 3: Manually enable the trunk link on a trunking port by using the **switchport mode trunk** command.

Step 4: Disable DTP (auto trunking) negotiations on trunking ports by using the **switchport nonegotiate** command.

Step 5: Set the native VLAN to a VLAN other than VLAN 1 by using the **switchport trunk native vlan *vlan_number*** command.

```
S1(config)# interface range fa0/21 - 24
S1(config-if-range)# switchport mode trunk
S1(config-if-range)# switchport nonegotiate
S1(config-if-range)# switchport trunk native vlan 999
S1(config-if-range)# end
S1#
```

11.3 Mitigate DHCP Attacks

DHCP Spoofing Attack

A DHCP spDHCP Spoofing Attack

A DHCP spoofing attack occurs when a rogue DHCP server is connected to the network and provides false IP configuration parameters to legitimate clients. A rogue server can provide a variety of misleading information:

- **Wrong default gateway** - The rogue server provides an invalid gateway or the IP address of its host to create a man-in-the-middle attack. This may go entirely undetected as the intruder intercepts the data flow through the network.
- **Wrong DNS server** - The rogue server provides an incorrect DNS server address pointing the user to a nefarious website.
- **Wrong IP address** - The rogue server provides an invalid IP address effectively creating a DoS attack on the DHCP client.

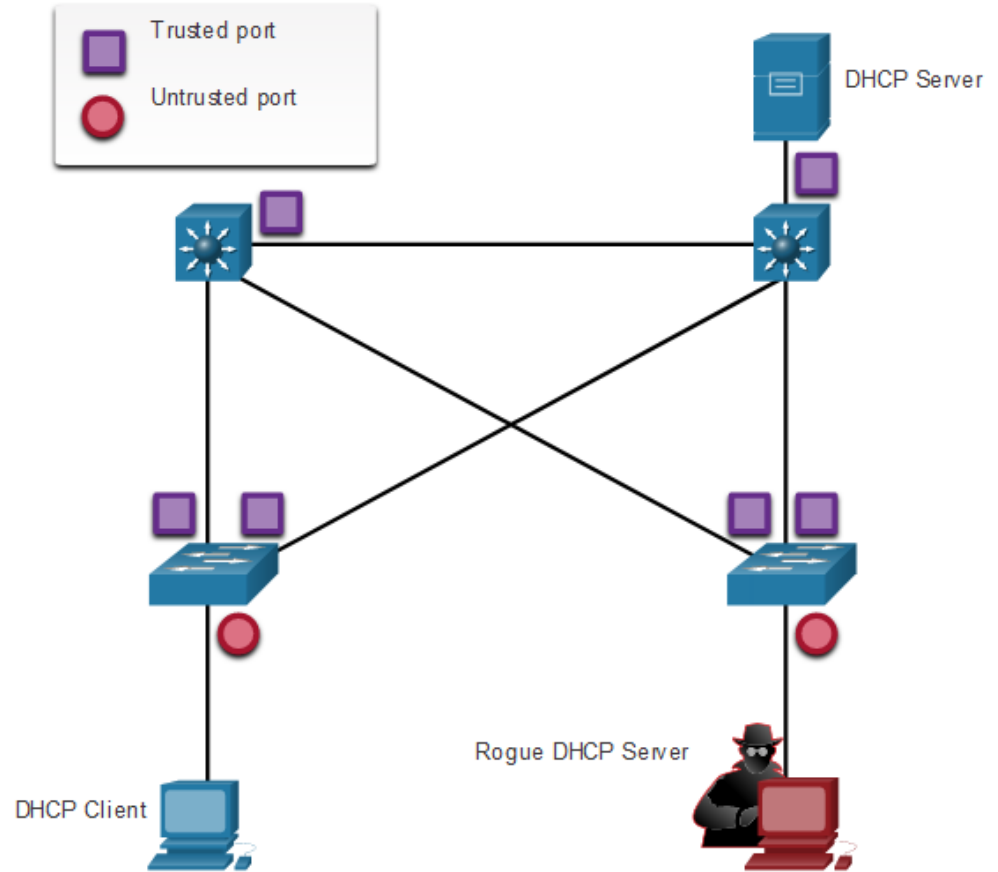
Mitigate DHCP Attacks

DHCP Attack Review

DHCP spoofing attacks requires more protection.

Port security is ineffective because the source MAC address would be legitimate.

DHCP spoofing attacks can be mitigated by using DHCP snooping on trusted ports.



Mitigate DHCP Attacks

DHCP Snooping

DHCP snooping

- **filters** DHCP messages
- **rate-limits** DHCP traffic on untrusted ports.
- Devices under administrative control (e.g., switches, routers, and servers) are trusted sources.
- Trusted interfaces (e.g., trunk links, server ports) must be explicitly configured as trusted.
- Devices outside the network and all access ports are generally treated as untrusted sources.

A DHCP table is built that includes the source MAC address of a device on an untrusted port and the IP address assigned by the DHCP server to that device.

- The MAC address and IP address are bound together.
- Therefore, this table is called the DHCP snooping binding table.

Steps to Implement DHCP Snooping

Use the following steps to enable DHCP snooping:

Step 1. Enable DHCP snooping by using the

Switch(Config)# ip dhcp snooping

Step 2. On trusted ports, use the

Switch(config-if)# ip dhcp snooping trust .

Step 3: On untrusted interfaces, limit the number of DHCP discovery messages that can be received using the

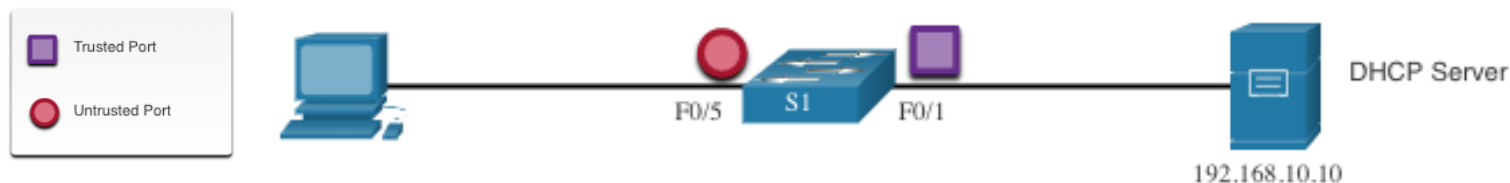
Switch(config-if)# ip dhcp snooping limit rate *packets-per-second*

Step 4. Enable DHCP snooping by VLAN, or by a range of VLANs, by using the

Switch(config)# ip dhcp snooping *vlan*

DHCP Snooping Configuration Example

Refer to the DHCP snooping sample topology with trusted and untrusted ports.



- DHCP snooping is first enabled on S1.
- The upstream interface to the DHCP server is explicitly trusted.
- F0/5 to F0/24 are untrusted and are, therefore, rate limited to six packets per second.
- Finally, DHCP snooping is enabled on VLANS 5, 10, 50, 51, and 52.

```
S1(config)# ip dhcp snooping
S1(config)# interface f0/1
S1(config-if)# ip dhcp snooping trust
S1(config-if)# exit
S1(config)# interface range f0/5 - 24
S1(config-if-range)# ip dhcp snooping limit rate 6
S1(config-if-range)# exit
S1(config)# ip dhcp snooping vlan 5,10,50-52
S1(config)# end
S1#
```

DHCP Snooping Configuration Example (Cont.)

Use the **show ip dhcp snooping** privileged EXEC command to verify DHCP snooping settings.

Use the **show ip dhcp snooping binding** command to view the clients that have received DHCP information.

Note: DHCP snooping is also required by Dynamic ARP Inspection (DAI).

```

S1# show ip dhcp snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
5,10,50-52
DHCP snooping is operational on following VLANs:
none
DHCP snooping is configured on the following L3 Interfaces:
Insertion of option 82 is enabled
  circuit-id default format: vlan-mod-port
  remote-id: 0cd9.96d2.3f80 (MAC)
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Verification of giaddr field is enabled
DHCP snooping trust/rate is configured on the following Interfaces:

```

Interface	Trusted	Allow option	Rate limit (pps)
FastEthernet0/1	yes	yes	unlimited
FastEthernet0/5	no	no	6
FastEthernet0/6	no	no	6

```

Custom circuit-ids:
Custom circuit-ids:
S1# show ip dhcp snooping binding

```

MacAddress	IpAddress	Lease(sec)	Type	VLAN	Interface
00:03:47:B5:9F:AD	192.168.10.10	193185	dhcp-snooping	5	FastEthernet0/5